

Inequalities Solutions

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§1 Walkthrough Problems

Problem 1.1. If $abcd = 1$ for $a, b, c, d > 0$, prove that

$$a^4b + b^4c + c^4d + d^4a \geq a + b + c + d.$$

Proof. By Hölder, we have:

$$\begin{aligned} \left(\sum_{\text{cyc}} a^4b \right) \left(\sum_{\text{cyc}} a \right) \left(\sum_{\text{cyc}} c \right) \left(\sum_{\text{cyc}} d \right) &\geq \left(\sum_{\text{cyc}} \sqrt[4]{a^4abcd} \right)^4 \\ \implies \left(\sum_{\text{cyc}} a^4b \right) (a + b + c + d)^3 &\geq (a + b + c + d)^4 \end{aligned}$$

And we're done. □

Problem 1.2 (IMO 2001). Let a, b, c be positive reals. Prove that

$$\frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ca}} + \frac{c}{\sqrt{c^2 + 8ab}} \geq 1.$$

Proof. Use Hölder to get the following:

$$\begin{aligned} \left(\sum_{\text{cyc}} \frac{a}{\sqrt{a^2 + 8bc}} \right)^2 \left(\sum_{\text{cyc}} a(a^2 + 8bc) \right) &\geq \left(\sum_{\text{cyc}} \sqrt[3]{\frac{a^2}{a^2 + 8bc} \cdot a(a^2 + 8bc)} \right)^3 \\ &= \left(\sum_{\text{cyc}} a \right)^3 \end{aligned}$$

However we note:

$$\left(\sum_{\text{cyc}} a \right)^3 = (a + b + c)^3 \geq a^3 + b^3 + c^3 + 24abc = \left(\sum_{\text{cyc}} a(a^2 + 8bc) \right)$$

So we're done. □

Problem 1.3 (IMO Shortlist 2009). Let a, b, c be positive real numbers such that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = a + b + c$. Prove that:

$$\frac{1}{(2a + b + c)^2} + \frac{1}{(a + 2b + c)^2} + \frac{1}{(aa + b + 2c)^2} \leq \frac{3}{16}.$$

Proof. We begin by rewriting.

$$\begin{aligned}\sum_{\text{cyc}} \frac{1}{(2a+b+c)^2} &\leq \frac{3}{16} \\ \sum_{\text{cyc}} \frac{16}{(2a+b+c)^2} &\leq 3 \\ \sum_{\text{cyc}} \frac{1}{a(a+b+c)} = 1 &\implies \sum_{\text{cyc}} \left(\frac{16}{(2a+b+c)^2} - \frac{3}{a(a+b+c)} \right) \leq 0\end{aligned}$$

Which means the inequality is homogenized. WLOG let $a+b+c=3$, then we have

$$\sum_{\text{cyc}} \left(\frac{16}{(3+a)^2} - \frac{1}{a} \right) \leq 0$$

$$\text{Define } f(x) = \frac{16}{(x+3)^2} - \frac{1}{x} \text{ then compute } f''(x) = \frac{96}{(x+3)^4} - \frac{2}{x^3}.$$

Note that f is concave on $[0, 3]$. Therefore, by Jensen,

$$\frac{f(a) + f(b) + f(c)}{3} \leq 3f\left(\frac{a+b+c}{3}\right) = 3f(1) = 0 \implies \boxed{f(a) + f(b) + f(c) \leq 0}.$$

□

§2 Problems

Problem 2.1 (Canadian Olympiad 2002). Let a, b, c be positive reals. Prove that

$$\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} \geq a + b + c.$$

Proof. By AM-GM, we have:

$$\frac{a^3}{bc} + b + c \geq 3a$$

and summing cyclically and simplifying gives our result. □